UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER EXAMINATION

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES

TUESDAY, 1 OCTOBER 2024 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATION AND NETWORKING
BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGU
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **CHOOSE ANY ONE (1)** question in **Section B**. Each question carries 25 marks each.

Note: Should you answer more than ONE (1) question in section B, marks will only be awarded to the FIRST question attempted.

SECTION A (Answer All Questions)

Q1. (a) Read the following scenario:

An online bookstore wants to design and implement a new system that allows customers to search for books, view details about the books, add books to a shopping cart, and purchase books online. The system should also support user authentication, order processing, and inventory management.

Referring to the above scenario, answer the following questions.

(i) Propose **THREE (3)** functional requirements for the online bookstore system. (6 marks)

(ii) Propose **THREE (3)** non-functional requirements for the system. (6 marks)

(b) Draw a concept map illustrating the incremental development model by using the scenario in Q1 (a). (5 marks)

(c) Explain **TWO (2)** benefits and **TWO (2)** challenges associated with reuse-oriented software engineering. (8 marks)

[Total : 25 marks]

Q2. (a) Discuss **THREE (3)** key roles in a Scrum team. (6 marks)

(b) (i) Describe **FOUR (4)** main components of a Sprint Backlog. (4 marks)

(ii) Based on Q2 (b)(i), explain how Sprint Backlog is created throughout the Sprint. (10 marks)

(c) In consideration of designing an Airline ticket booking system, you study and analyzed several existing mobile applications in the current market. You decided to develop the system with the following functionalities.

• Firstly, a user enables to search for desired airline ticket based on users' preferences through online booking or able to directly purchase the ticket.

• Secondly, the user will receive a notification for the travel date and time of his/her airline.

Q2. (c)(Continued)

(i) Assuming you are the user, provide **FIVE (5)** user stories to express the functional requirements, excluding user registration, sign in and sign out. (5 marks)

[Total : 25 marks]

Q3. (a) Draw diagrams to show conceptual view and the process view of the architectures of the following systems:

(i) An automated ticket-issuing system used by passengers at a railway station. (4 marks)

(ii) A computer-controlled video conferencing system that allows video, audio, and computer data to be visible to several participants at the same time. (4 marks)

(b) LKY is requested to design a learning management system. This system aims to record raw data of student's assessment results and generate statistics for students and lecturer. By using this system, lecturer can monitor students learning effectiveness. Therefore, it is very important to develop visualization functionality for presenting different charts of raw data so that a lecturer can view the students learning outcome in a bar charts, pie charts, curve diagrams, etc.
Based on the requirement of visualization in this learning management system, please answer the following questions:

(i) Suggest the most appropriate "Architectural pattern" to develop the visualization tool and explain **TWO (2)** conditions when this architectural pattern can be used. (5 marks)

(ii) Based on Q3(b)(i), describe the components or elements of this pattern. (6 marks)

(c) Discuss **THREE (3)** problems of application system integration. (6 marks)

[Total : 25 marks]

SECTION B (Choose Any One Questions)

Q4. (a) A medium-sized software development company is considering the adoption of Commercial-off-the-shelf (COTS) components to accelerate their development process and reduce costs.
As a software engineer, you are tasked with evaluating the potential questions below;

(i) Determine what Commercial-off-the-shelf (COTS) products in term of COTS reuse. (2 marks)

(ii) Discuss which type of Commercial-off-the-shelf (COTS) reuse is more appropriate? Justify your choice. (4 marks)

(iii) Analyze **TWO (2)** risks associated with the adoption of Commercial-off-the-shelf (COTS) reuse. (4 marks)

(b) Analyze how software reuse through COTS components can positively impact the maintainability and scalability of a software project. (5 marks)

(c) Analyze **FIVE (5)** key steps involved in performing a maintenance task in software engineering. Your answer should include an explanation of each step and its importance in ensuring the software system remains functional and efficient. (10 marks)

[Total : 25 marks]

Q5. (a) Analyze and explain **TWO (2)** types of knowledge that might be captured in organizational standards. (5 marks)

(b) Software is increasingly being developed by teams where the team members are working at different locations. Discuss **FIVE (5)** essential features that a version management system should support to facilitate distributed software development. (10 marks)

Q5. (Continued)

(c) Assume that serial numbers of an identification card (IC) can only accept the valid auto-generated values with the first and second criterion as follows.

- Criterion 1: The array of valid random values shall be between "870119085551" and "870119-08-5551.
- Criterion 2: The sequence of "year", "month", "birth date", "state code", and random four numbers only must be fulfilled
  It applies "tolower()" function in C++ programming.

According to the requirements above, answer the following questions using Guideline Based Testing:

(i) Draw an equivalence partition table for Criterion 1. (5 marks)

(ii) Provide **FIVE (5)** test cases for Criterion 2. Each test case must be written in a sentence. (5 marks)

[Total : 25 marks]
